

Entities

Components which support the proper functionality in a system is known as entities.

Nouns often represent potential entities.

eg:- real world object, abstract-concept, role, event, location

Entities are represented as a "Rectangle" in ERDs. Nouns inside them should be written in "Singular" form.

Relationships

Relationships are used to depict how the entities are connected to each other.

Relationships are represented as "Diamonds" between entities.

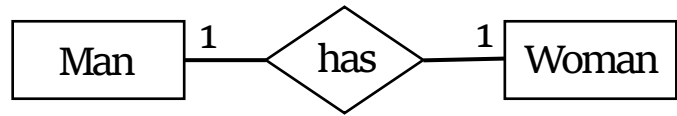
Most of the times, a relationship is denoted by a verb in a context.

Defining the cardinality in a relationship & specifying how many instances of one entity are related to instances of another is important.

It's recommended to use an appropriate notation such as Crow's foot or Chen Notation to represent the relationships visually.

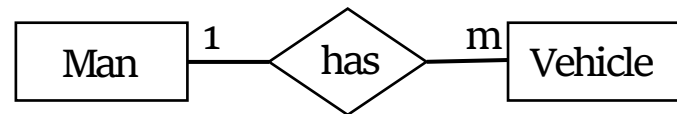
One to one relationship

One instance of an entity is allowed to relate with only one instance of another entity.



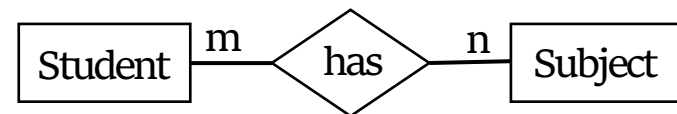
One to many relationship

One instance of an entity is allowed to relate with more than one instance of another entity. But one instance of the 2nd entity is only allowed to connect with one instance of the 1st entity.



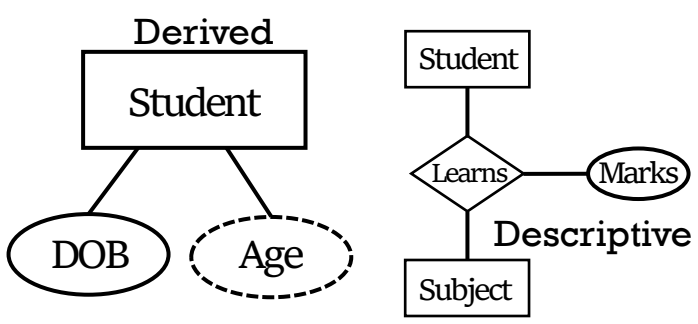
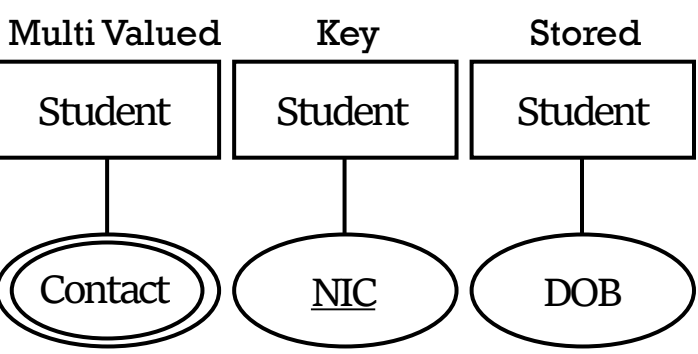
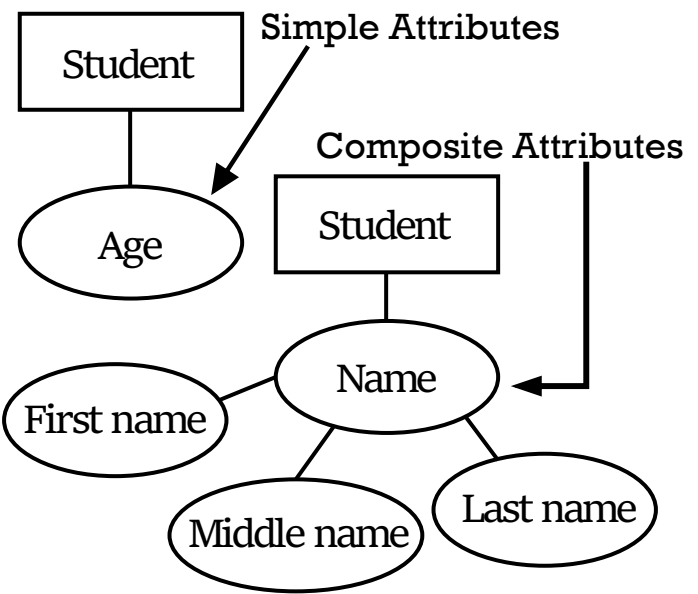
Many to many relationship

Any number of instances in one entity are allowed to connect with any number of instances of the other entity. (no restrictions)

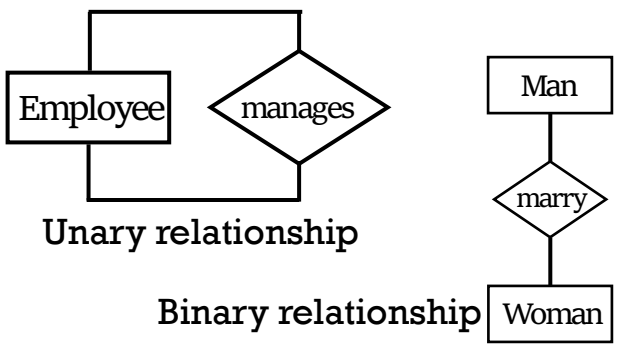


Attributes

These are the properties that describe the entities & are represented using an "Oval" shape. For each entity, there is a list of characteristics which defines it. These become the attributes in an ERD. Some relationships may have attributes that describe the properties of the relationship itself. Identifying them is also crucial.



Degree of a relationship

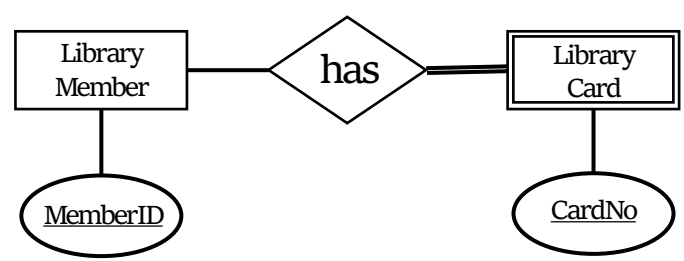


Strong Entities

It can exists independently from other entities. It has a “primary key” to uniquely identify the each instance of the entity. These can be uniquely identified by its attributes alone. Strong entities are usually represented by a normal rectangle.

Weak Entities

These can not exist without a relationship with a strong entity. It doesn't have a primary key to uniquely identify the each instance of the entity & cannot be uniquely identified by its attributes alone. It relies on a combination of its own attributes and the primary key attributes of the related strong entity. Weak entities typically represent dependent parts or components of a strong entity. Usually displayed with a double rectangle in an ERD.



Strong & Weak relationships

These share the same energy as strong & weak entities. A weak relationship forms when a weak entity is connected to a strong entity.



Weak Participation

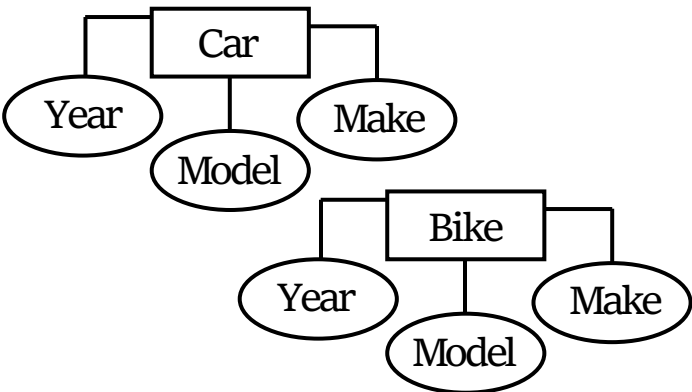
Every instance of the entity is not required to participate in a relationship. Partial participation is displayed by a single line connecting the entity and the relationship.

Total Participation

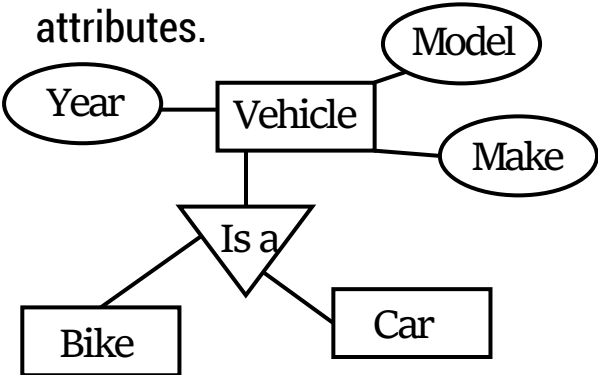
Every instance of the entity must participate in the relationship. Total participation is displayed by double lines connecting the entity and the relationship.

Generalization

Generalization is the process of extracting common properties from multiple entities to create a more general entity. It's useful when different entities share common attributes or behavior, and there's a need to eliminate redundancy.



We can generalize these two into a more general entity called "Vehicle", which encapsulates the common attributes.



Specialization

Specialization is the opposite of generalization. It's the process of creating new entities from an existing entity based on some distinguishing characteristics. Ideal when an entity can be divided into more specific categories, each having unique attributes or relationships. Typically shown by a higher-level entity with the specialized ones. This intends to highlight unique characteristics.

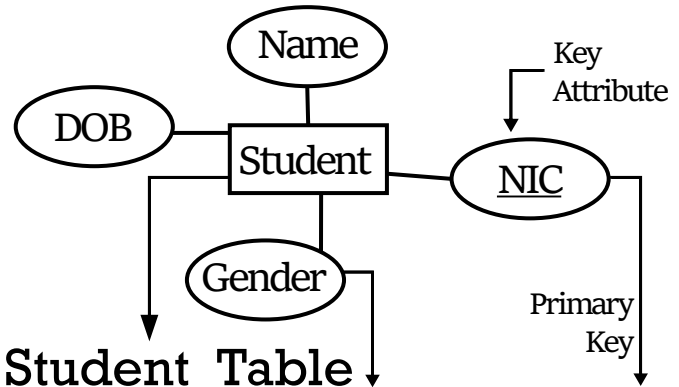
Sub Class

A subclass is a specialized entity that inherits attributes and relationships from a more general entity.

Super Class

A superclass is a more general entity that encapsulates the common attributes and relationships shared by its subclass.

Mapping ERDs into relational Database Tables



Student Table

Name	DOB	Gender	<u>NIC</u>

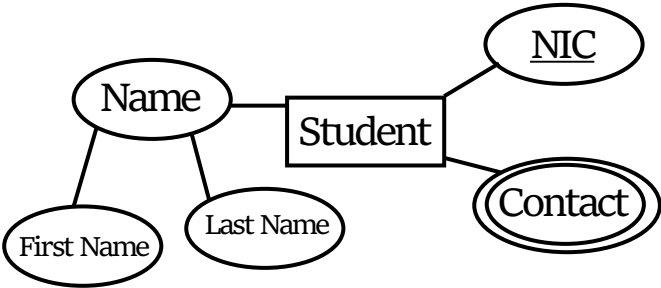
Mapping Composite, Derived, Multi Valued attributes

Composite attributes are depicted each part as a separate column of the table, with representing many attributes.

Derived attributes are not represented in database tables. Instead, they are calculated from the stored attributes when needed.

Multi-Valued attributes should be displayed in a separate table, with a copy of the key attribute of the main table.

Mapping descriptive attributes will be discussed after mapping relationships.



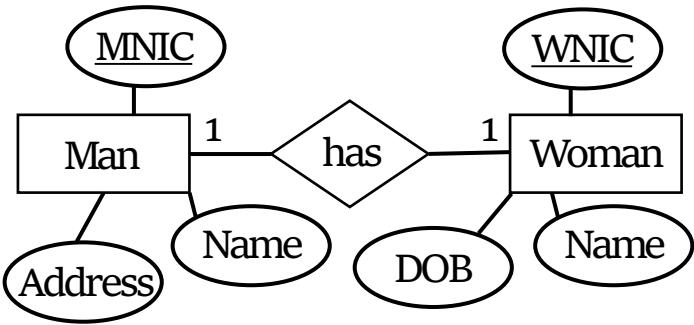
Student_Table

<u>NIC</u>	First Name	Last Name

Contact_Table

<u>NIC</u>	Contact

Mapping one to one relationships



Man_Table (MNIC, Name, DOB)
Marriage_Table (MNIC, WNIC)
Woman_Table (WNIC, Name, DOB)

Man_Table

<u>MNIC</u>	Name	DOB

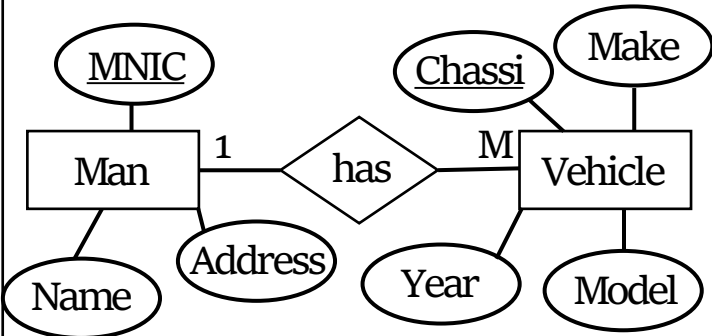
Woman_Table

<u>WNIC</u>	Name	DOB

Marriage_Table

<u>MNIC</u>	<u>WNIC</u>

Mapping one to many relationships



Man_Table (NIC, Name, Address)
Vehicle_Table (Chassi, Make, Model, Year, NIC)

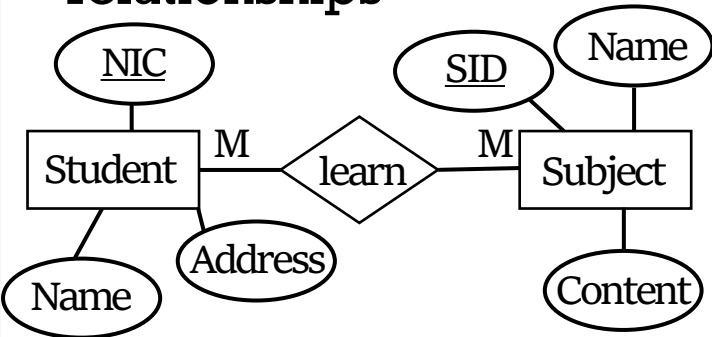
Man_Table

<u>NIC</u>	Name	Address

Vehicle_Table

<u>Chassi</u>	Make	Model	Year	NIC

Mapping many to many relationships



Student_Table (NIC, Name, Address)
Learn_Table (NIC, SID)
Woman_Table (SID, Name, Content)

Student_Table

<u>NIC</u>	Name	Address

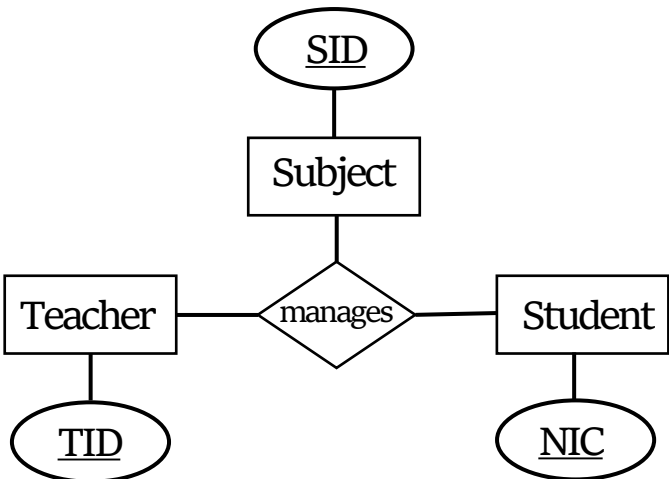
Marriage_Table

<u>NIC</u>	<u>SID</u>

Subject_Table

<u>SID</u>	Name	Content

Mapping ternary relationships



Create a new table. This table represents the ternary relationship.

Include the primary keys from each of the three related entity table as foreign keys in this new table.

If the ternary relationship has any descriptive attributes, also add them as columns in this table.

Often, the composite of the three foreign keys will be used as the primary key of this table.

Student_Table

<u>NIC</u>	XXXX	XXXX	XXXX

Teacher_Table

<u>TID</u>	XXXX	XXXX	XXXX

Subject_Table

<u>SID</u>	XXXX	XXXX	XXXX

Relationship_Table

NIC	TID	SID	Descriptive attributes

Mapping Descriptive Attributes into Tables

Binary one to one

Descriptive attributes can be added in either tables, or even in a separate table linked via foreign keys.

Binary one to many

Descriptive attributes should be added to the "many" sides table or even in a separate table linked via foreign keys.

Binary many to many

Descriptive attributes can be added to the relationship table to provide additional information about the relationship.

Unary Relationships

One to one and one to many, a descriptive attribute can be added directly to the table as a separate column. In many to many, they should be added to the separate junction table containing two columns with the primary key of the entity.

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Ternary Relationships

The descriptive attributes are included in the separate linking table containing foreign keys to each participating entity table.

eg :- In a relationship between "Suppliers", "Products" and "Stores", the relationship table includes attributes like "Delivery Date"

Super Key

A key which is made by combining a primary key field of a table with non-primary key of a table for better use is known as a super key.